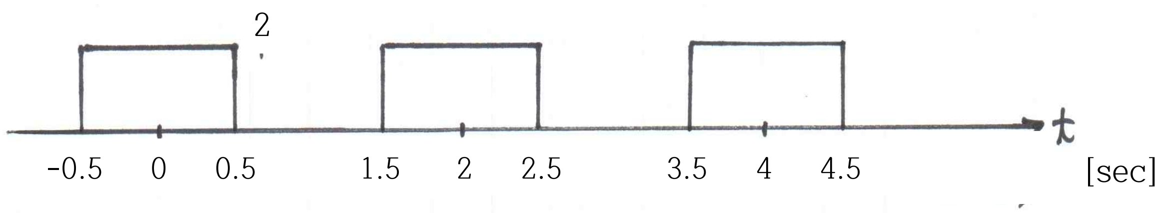
Homework 07

A periodic square wave is given. (**<u>Period T = 2 sec, Amplitude = 2</u>**)



1. Find ω_0 [rad/sec].

   (1) 0.5 (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{2}$ (4) 0.25 (5) π

2. Find the Fourier coefficient, a_k .

   (1) $\frac{1}{k\pi}\sin(\frac{k\pi}{2})$ (2) $\frac{4}{k\pi}\sin(\frac{k\pi}{4})$ (3) $\frac{2}{k\pi}\sin(\frac{k\pi}{2})$ (4) $\frac{4}{k\pi}\sin(\frac{k\pi}{2})$ (5) $\frac{2}{k\pi}\sin(\frac{k\pi}{4})$

3. Find the first zero crossing frequency in Hz?

   (1) 0 (2) 0.5 (3) 1 (4) 1.5 (5) 2

4. Find the frequency in Hz of the spectrum interval?

   (1) 1 (2) 0.5 (3) 0.25 (4) 1.5 (5) 0

5. Find the average value of this signal?

   (1) 0 (2) 0.5 (3) -0.5 (4) 1 (5) -1

6. Find a_{-2} .

   (1) $-\frac{1}{\pi}$ (2) $-\frac{2}{\pi}$ (3) 0 (4) $\frac{2}{\pi}$ (5) $\frac{1}{\pi}$

7. Find a_{-1} .

   (1) $-\frac{4}{\pi}$ (2) $\frac{1}{\pi}$ (3) $-\frac{2\sqrt{2}}{\pi}$ (4) $\frac{\sqrt{2}}{\pi}$ (5) $\frac{2}{\pi}$

8. Find a_0 .

   (1) 0 (2) $\frac{1}{2}$ (3) $\frac{2}{\pi}$ (4) $\frac{1}{\pi}$ (5) 1

9. Find a_1 .

   (1) $\frac{\sqrt{2}}{\pi}$ (2) $\frac{2}{\pi}$ (3) $-\frac{1}{\pi}$ (4) $\frac{2\sqrt{2}}{\pi}$ (5) $\frac{1}{\pi}$

10. Find a_2 .

    (1) 0 (2) $-\frac{1}{\pi}$ (3) $\frac{1}{\pi}$ (4) $-\frac{1}{2\pi}$ (5) $\frac{2}{\pi}$

1	2	3	4	5	6	7	8	9	10